

PALAMAKULA DEEPIKA

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Professional Summary

I am eager to join this organization to contribute to the future of sustainable transport and infrastructure. My goal is to apply my expertise in Data Engineering (ETL), Statistical Mathematics, and Power BI to support Data-Driven Powertrain Development. I am committed to delivering high-impact solutions within Agile teams, moving models from initial script to test cell verification, while continuously evolving my skills in Generative AI and Predictive Analytics to drive global industrial innovation.

Education

Bachelors of technology in Artificial Intelligence & Machine Learning | 2021 – 2025

Saveetha Engineering College | Chennai, TamilNadu, India | CGPA 8.2/10.0

Technical Skills

- **Data Engineering & Analytics:** ETL Automation, Statistical Mathematics, Data Structures & Algorithms, Pandas, NumPy, Scikit-learn, XGBoost, Feature Engineering.
- **Generative AI & Deep Learning:** Researching LLM applications, Neural Networks, Deep Learning.
- **Data Visualization:** Power BI, Interactive Dashboards, Microsoft Excel, SHAP (Explainable AI).
- **Cloud & DevOps:** AWS (EC2, S3), Git/GitHub, MLOps Pipelines, Google Cloud (GCP).
- **Programming & IoT:** Python, Java, C, Arduino IDE, ESP32/Sensor Integration, SQL (Oracle).
- **Core Competencies:** Model Deployment, Feature Engineering, Data-Driven Decision Making.

Work Experience

Machine Learning Intern | FabHost Complete Web Sol. | Jan 2024 – Mar 2024

- Developed and automated ETL scripts using Python to streamline the ML Lifecycle, reducing manual data preparation time by 30%.
- Engineered Supervised ML models (XGBoost) for predictive challenges, achieving a 92% Precision/Recall score through rigorous model validation and SHAP analysis.
- Automated data cleaning and integration processes by developing custom Python scripts, significantly reducing manual data preparation time and streamlining the ML Lifecycle (MLOps).

Projects

CareerLens AI: Scalable Career Recommendation System | Python, FastAPI, Flutter, AWS

- Architected a 3-layer AI system (Flutter, FastAPI, Python ML) to provide personalized career path recommendations and skill gap analysis with 90%+ predictive accuracy.
- Integrated Explainable AI (SHAP) with RandomForest/XGBoost models to offer transparent, feature-level reasoning and confidence scores for user recommendations.
- Deployed a cloud-ready infrastructure on AWS EC2 and S3 using MongoDB for scalable data storage, ensuring modular design and seamless RESTful API communication.

Toxic Gas Leak Detection & Alerting Device | IoT, ESP32, Random Forest, IFTTT

- Developed an IoT safety system utilizing ESP32 and MQ135 sensors to monitor atmospheric toxicity and deliver real-time SMS/Email alerts via IFTTT protocols.
- Implemented Random Forest Regression (RFR) to analyze sensor telemetry and predict hazard levels, reducing potential emergency response time through automated detection.
- Engineered a cloud-integrated monitoring solution using Arduino IDE and Blynk, enabling remote device control and live data visualization of environmental metrics.

Certifications

- Data Structures & Algorithms (C & Java) | Udemy | 2023
- Fundamentals of DevOps | Simplilearn | 2024
- AWS Certified Cloud Practitioner | Amazon Web Services (AWS) | 2025